# STRUCTURES

# LEARNING OUTCOMES

At the end of the session, students are expected to:

1. Define what a structure is
2. Differentiate a structure from an array
3. Declare a structure type
4. Declare and use structure variables
5. Apply nested structures and arrays within structures in coding

# WHAT IS A STRUCTURE?

A structure is a user-defined data type that allows you to group related variables of different data types under one name.

It is used when a single entity needs multiple pieces of information.

```
struct Student
{
    public int Id;
    public string Name;
    public double Grade;
}
```

```
Student s1;
s1.Id = 1;
s1.Name = "Juan";
s1.Grade = 90.5;

Console.WriteLine(s1.Name);
```

# STRUCTURE VS ARRAY

| Aspect | Structure | Array |
|---|---|---|
| Data Types | Can store **different** data types | Stores **only one** data type |
| Purpose | Represents **one complete entity** | Represents **many similar values** |
| Access | By **member name** | By **index number** |
| Flexibility | High (complex data modeling) | Limited |
| Example Use | Student, Employee, Patient | List of scores, prices |

# STRUCTURE WITHIN A STRUCTURE

A structure within a structure means one structure is used as a member of another structure.

This is helpful when data itself has sub-components.

Why Use Nested Structures?

- Makes data more organized
- Matches real-world relationships
- Easier to maintain and expand

```
struct Address
{
    public string City;
    public int ZipCode;
}

struct Student
{
    public int Id;
    public string Name;
    public Address Addr;
}
```

```
Student s1;
s1.Id = 1;
s1.Name = "Ana";
s1.Addr.City = "Cebu";
s1.Addr.ZipCode = 6000;

Console.WriteLine(s1.Addr.City);
```

# WHEN TO USE STRUCTURE WITHIN A STRUCTURE

- When data has logical subgroups
- When the same sub-structure can be reused
- When modeling real-world entities

Examples:

- Employee → Department
- Patient → MedicalRecord
- Order → CustomerDetails

# ARRAY WITHIN A STRUCTURE

An array within a structure means a structure contains multiple values of the same type.

Why Use an Array Inside a Structure?

- To store repeated information for one entity
- To avoid creating multiple separate variables

```
struct Student
{
    public int Id;
    public string Name;
    public int[] Grades;
}
```

```
Student s1;
s1.Id = 1;
s1.Name = "Mark";
s1.Grades = new int[3] { 90, 85, 88 };

Console.WriteLine(s1.Grades[0]);
```

# STRUCTURE WITH ARRAY VS ARRAY OF STRUCTURES

Structure with Array

```
struct Student
{
    public int Id;
    public string Name;
    public int[] Grades;
}
```

One student → many grades

Array of Structures

```
struct Student
{
    public int Id;
    public float Grade;
}

Student[] students = new Student[3];

students[0].Id = 1;
students[0].Grade = 90.5f;
```

Many students → each with their own data

# THANK YOU